

Phase 3: Project Design Phase

Problem-Solution Fit

1. Value Proposition Canvas

The Problem-Solution Fit analysis confirms that the features engineered into ComicCraft directly address the jobs-to-be-done, alleviate the core pains, and maximize the gains of our target users.

THE PRODUCT (ComicCraft)		THE CUSTOMER (Storyteller)	
[Gain Creators]		[Customer Gains]	
- 1-Click 5-panel comic generation		- Create graphic novels without drawing	
- Print-ready PDF download with 1 page/panel		- Tangible, shareable finished publication	
- Style & Tone customization options		- Full creative control over mood and genre	
[Pain Relievers]		[Customer Pains]	
- Automated AI illustration inking		- High cost of human illustrators	
- Dual LLM orchestration (Flash + Pro)		- Incoherent story beats between images	
- Automatic typography & layout binding		- Tedious graphic design software learning	
[Products & Services]		[Customer Jobs]	
- Web GUI & REST API		- Conceptualize story plot & characters	
- Synchronized speech bubbles & narrations		- Produce visual storyboards for education	
- Offline PDF export module		- Share creative works with an audience	

2. Direct Feature-to-Problem Mapping

User Need / Pain Point	Traditional Obstacle	ComicCraft Engineered Solution
Cohesive Sequential Plot	Generic image tools generate single isolated pictures with no chronological plot progression.	Gemini 1.5 Flash Outline Generator: Enforces a strict 5-panel dramatic arc (Threshold -> Rising Action -> Confrontation -> Climax -> Resolution).
Authentic Comic Voice	Text descriptions lack character personality and dialogue pacing.	Gemini 1.5 Pro Narrative Expander: Writes distinct narrator lines (NARRATION:), atmospheric environment tags (CAPTION:), and quoted character speech.
Artistic Visualization	Hand-drawing requires decades of skill; commercial tools require complicated prompting.	Multi-Model Image Generator: Automatically embeds user-selected art style (Anime, Pixel Art, Comic Book, Realistic) into detailed visual scene prompts.
Publishing / Output	Exporting a comic requires manual page layout, margin adjustment, and font embedding.	FPDF Layout Exporter: Automatically renders 5 standardized A4 pages, centering illustrations and styling text in milliseconds.

3. Validation Criteria & Success Metrics

- 1. Creation Velocity: Average time from user form submission to PDF download readiness is under 15 seconds.
- 2. Sequential Cohesion: 100% of generated comics follow a continuous narrative arc across all 5 panels.
- 3. Format Integrity: Exported PDFs contain all 5 panel titles, illustrations, captions, and narrations on dedicated individual pages without clipping.